Project Initialization and Planning Phase

Date	02 July 2026
Team ID	
Project Title	A Comprehensive Measure of Well Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform human development assessment using machine learning, improving accuracy and insight into national well-being. It tackles the limitations of income-only development measures, promising a more holistic, data-driven classification and clearer guidance for policy intervention. Key features include a machine learning-based HDI prediction model and instant development-tier classification.

Project Overview	
Objective	The primary objective is to accurately measure and classify a country's level of human development by combining life expectancy, educational attainment, and income indicators into a single machine learning-based prediction model.
Scope	The project comprehensively evaluates a country's human development by processing health, education, and income data, and classifies the outcome into Very High, High, Medium, or Low development tiers, offering a broader perspective than income-based measures alone.
Problem Statement	
Description	Development is often assessed using economic growth or income alone, which does not fully capture a population's health, education, and overall quality of life, resulting in an incomplete picture of true human development.
Impact	Solving this issue will result in a more holistic, data-driven measure of development that helps governments, researchers, and international organizations identify gaps and prioritize interventions in healthcare, education, and income generation.
Proposed Solution	
Approach	Employing machine learning techniques to analyze life expectancy, mean years of schooling, expected years of schooling, and GNI per capita (PPP) together, creating a predictive system that classifies countries into human development tiers.
Key Features	- Implementation of a machine learning-based HDI prediction model. - Real-time classification into Very High, High, Medium, or Low development tiers. - Interactive input form for life expectancy, schooling, and income indicators.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard multi-core CPU

Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	50 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, PyCharm
Data		
Data	Source, size, format	UNDP Human Development Report dataset, country-level, csv